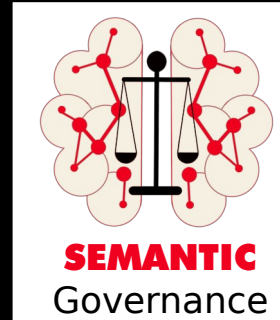


Semantic Governance

Conversational Search

Design and Initial Pilot Plan



Limitations of LLMs

Causal Reasoning Deficiency

- Struggle with causal reasoning. Can't reliably identify cause-and-effect relationships, crucial for making informed business decisions.

Quantitative Analysis

- Not setup to handle analytics and precise calculations let alone determining with digital assets to assemble to deliver the right precision in the right mode with the right focus & relevancy.

Decision-Making Support

- Fall short providing deep analytical insights necessary for strategic decisions based on quantitative data.

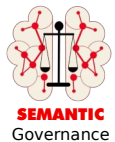
Integration Challenges

- Integrating LLMs into workflows that involve unstructured and structured data presents challenges.

In addition to these reasoning challenges, here are a few other areas to consider:

- Data and Bias - quality of LLM outputs is heavily influenced by the training data, which can contain biases.
- Hallucination - detrimental in decision-making as incorrect information can lead to unreliable conclusions.
- Explainability - lack explicit models to explain recommendations impacts understandability behind reasoning
- Transparency - poses challenges in validating and trusting AI-driven decisions.

Introducing Knowledge Graphs into RAG-Based Learning



Why do we need KGs with RAG?

- Vector databases (VDBs) excel in managing high-dimensional data, particularly useful for tasks involving embeddings, such as image and text similarity.
- Knowledge graphs (**KGs**), on the other hand, are designed to represent structured relationships between entities, making them more suitable for applications where understanding and navigating relationships (**explicit & tacit**) is critical.
- There's a growing emphasis on explainability in AI. KGs can provide clearer insights into how AI systems arrive at certain decisions

Why are companies using KGs?

- **KGs** are increasingly being used in scenarios requiring reasoning, inferencing, and complex queries about relationships, such as in natural language understanding, recommendation systems, and semantic search.

How are companies incorporating KGs into RAG?

- Some organizations are adopting hybrid approaches, utilizing both vector databases and knowledge graphs **where VDBs** handle similarity searches and **KGs** provide contextual relationships and metadata that enhance the understanding of the data.

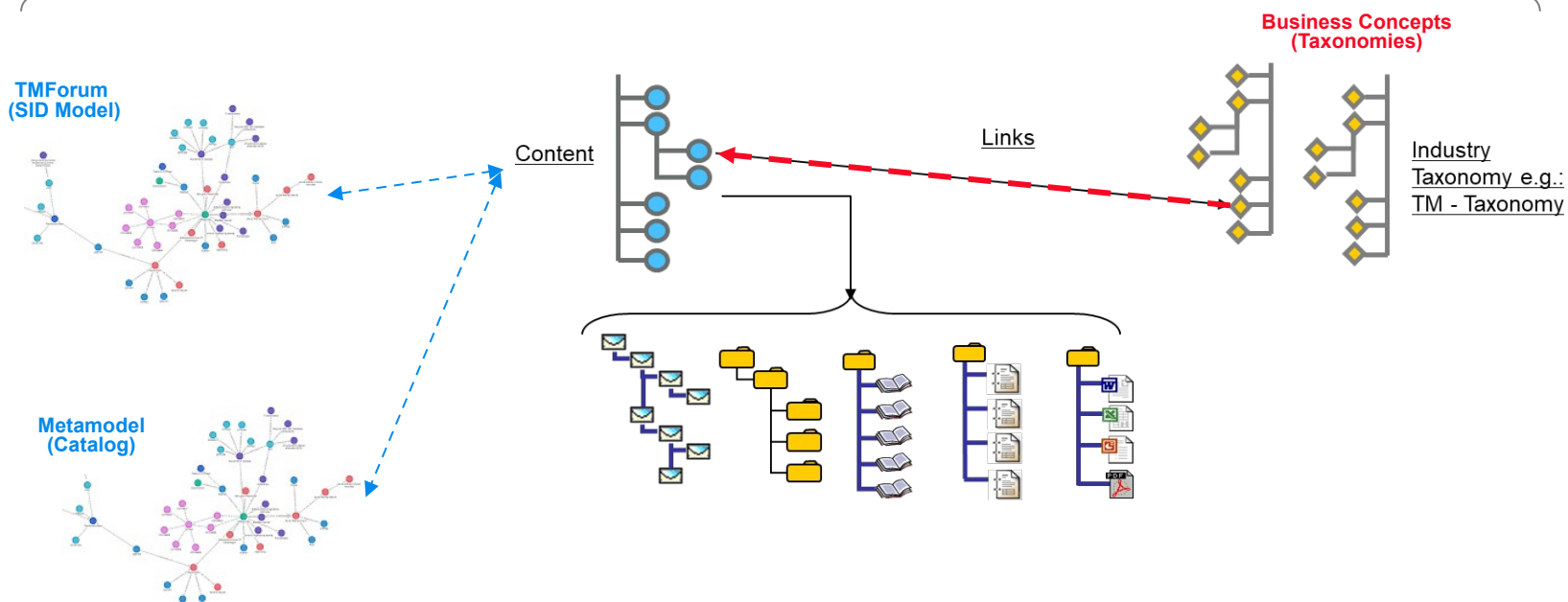
How well does this perform?

- **VDBs** are optimized for fast similarity searches and can handle large volumes of unstructured data
- **KGs** perform better when **presented** with queries that involve complex relationships.

Powering Enterprise Search

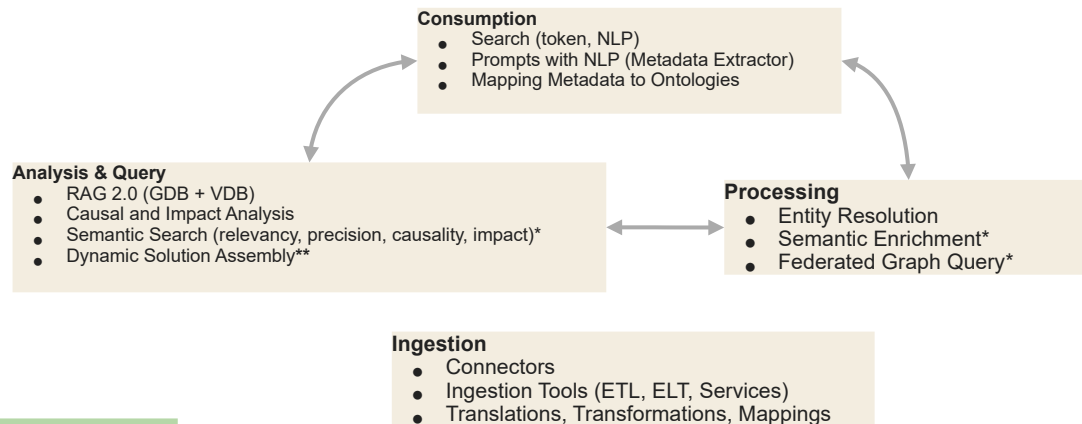
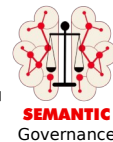


Business Oriented Search



Enabling Conversational Search

GDB = Graph DB, VDB = Vector DB



Entity Extraction: Use a graph transformer to extract entities from data structures, artifacts, documents, unstructured content and represent them as nodes in the relevant domain

Relationships Generation: Define relations (using standard relations defined in the Upper Ontology) between entities in the appropriate domain and/or across domains

*Value from GDB & Semantic Search

- This technology enables applications to synthesize data from various sources, offering a holistic view and uncovering hidden connections within datasets.
- The modular design of the system ensures scalability as the ontology and data grow.
- Using a structured data source such as a graph gives answers which are more deterministic and less dependent on the specific wording of the user query. This improves consistency and accuracy.

Persistence

- Federated Knowledge Graphs (Domain-Specific)
- Data Lineage & Data Quality
- Catalog (Metadata, Terms, Attributes)
- Policies & Controls (Provisioning, Protection)
- Asset Repositories (Data, Metrics, Models, Systems, Components)

Digital Asset Sources

- Alation (Calcs & Metrics)
- GCP Data Warehouse + EDW Warehouse + Hadoop Warehouse
- EnGRAM (AI Models) + Other AI Model repositories
- Media Stores, Document Stores
- Unstructured Content Stores (social media, emails, mined content,...)
- Lifecycle Repositories (GIT) Components, Systems, Artifacts

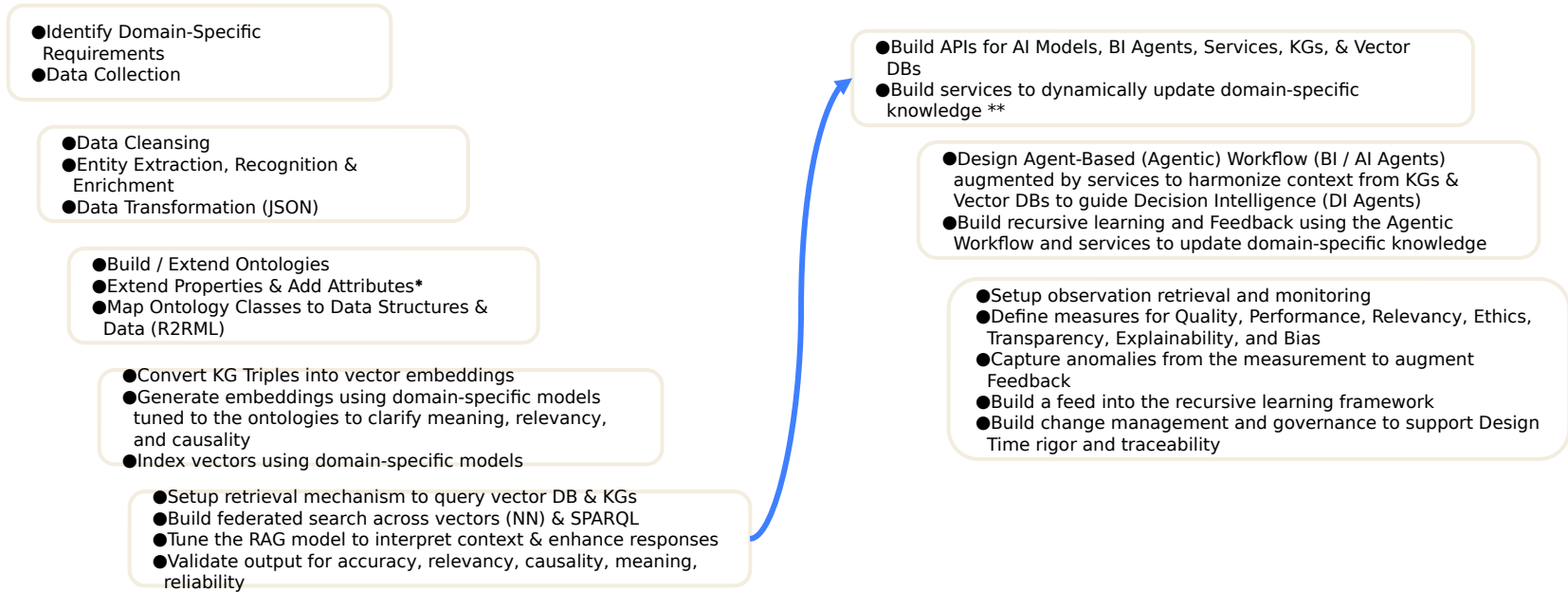
** Dynamic Solution Assembly

data + metrics, + dimensions + AI (relevant context for problem domains) + AI (personalization using relevant filtering & translation) + BI (appropriate visualization)

Validation & Quality: Refine the graph, remove duplicates, combining relations / edges, and consolidate nodes (carefully) to drive data consistency and clarity

Ontology Publishing: Once ontology changes are validated, publish new versions of ontologies and rebuild graphs for downstream consumption.

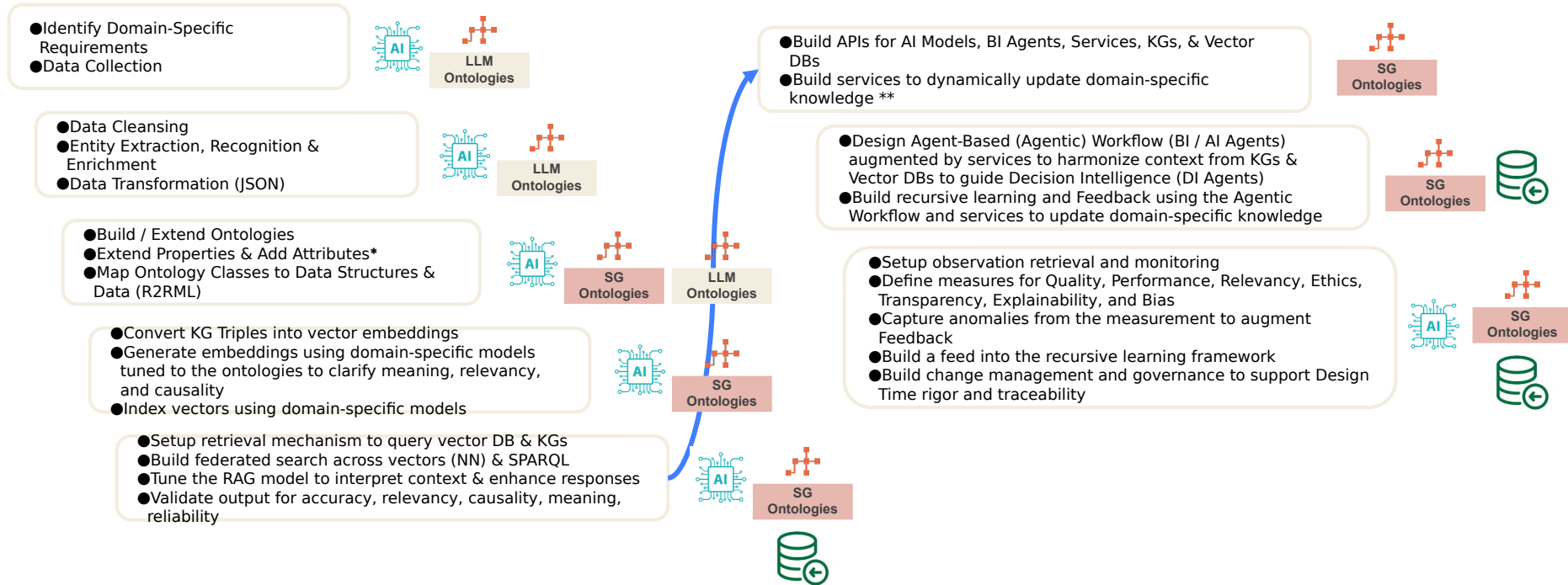
RAG-Based Learning Flow



*RDF-Star (future state standard supports
LPGs)

**Attributes, Relations (ontologies & vectors)

RAG-Based Learning Flow



*RDF-Star (future state standard supports
LPGs)

**Attributes, Relations (ontologies & vectors)

Enabling Conversational Exchange

Input

- Natural Language Prompt
- String & Phrase Parsing by LLM
- Clarify Relevancy using [Persona Dimensions](#)



LLM
Ontologies

Contextual Mapping

- Generate Domain Listing using Strings, Phrases, [Relevancy](#)
- Identify Domain-Specific Ontologies (short list of IRIs)*
- Generate Graph Queries using SPARQL + SHACL (constraints)*
- Obtain list of (Data) Sources for Data Extraction
- Obtain list of (Data) Entities using Class Mappings for Data Extraction
- Obtain list of (Data) Filters using constraints from SHACL
- Obtain list of (Data) Analytics (Metrics) & Presentation (visualization)



LLM
Ontologies



SG
Ontologies

* using **Metaphacts
APIs**

Data Extraction

- Use SPARQL to generate Relational Queries (SQL Sources + Data Entities)
- Use SHACL to generate constraints (Where Clauses & Conditional Logic)
- Send Relational Queries to GCP/BQ to generate one or more sets of extracts



LLM
Ontologies



Data Analysis & Presentation

- Use list of Metrics to execute Calcs against the data extracts using LookML
- Use list of Visualizations to provide the appropriate "Lens" for consumption



LLM
Ontologies



lookML



This is a very simple walkthrough to explain the process of capturing user input (question), parsing it into meaningful context, and building federated queries to extract the relevant content so it can be presented intuitively.

An actual conversation may require multiple loops between the first 2 steps to help clarify context.

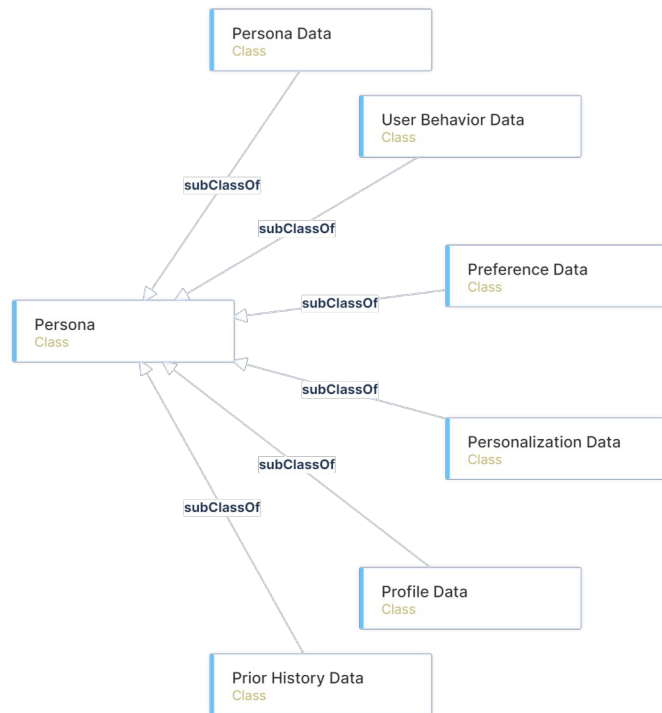
Mapping data transforms & metrics are simple for straight forward questions (linear and some distributed calcs).

Mapping the outputs into the visualizations (including a simple answer: a number) will take more time as there are several factors that drive the right presentation.

Appendix

Supporting Materials

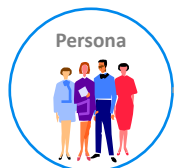
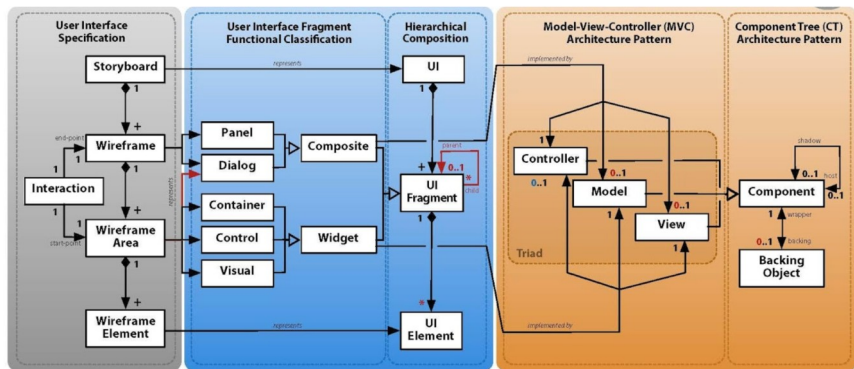
Driving Relevancy thru a Unified Persona Model (5 Dimensions)



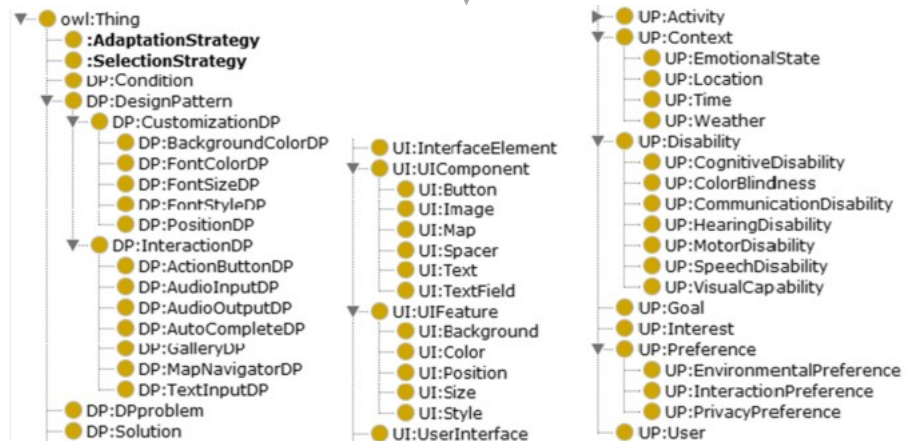
- 
- Provisioning
 - Protection
 - **Relevancy***
 - Notification
 - Escalation

*Push/Pull dynamic tailoring

Ontology Driven User Experience (adaptive & relevant)



Semantic
Mapping



Semantic Governance Integration

